THE COMPARISON MAP BETWEEN PULLBACK AND THE RIGHT ADJOINT IS AN ISOMORPHISM FOR A PROPER FLAT FINITE PRESENTATION MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0E4K (sources/stacks-duality/pinned-original-source.tex line 2892).

Allowed standard prerequisites (editorial).

- The right adjoint commutes with direct sums for a proper flat morphism. Same chapter, Lemma labelled lemma-proper-flat, cited at proof lines 2919-2921. Makes the isomorphism locus closed under direct sums.
- Generation of the quasi-coherent derived category by a single perfect object, in a module category. Derived Categories of Schemes, Theorems labelled perfect-theorem-DQCoh-is-Ddga and perfect-theorem-bondal-van-den-Bergh, cited at proof lines 2927-2933, and More on Algebra, Remark labelled more-algebra-remark-P-resolution, cited at proof lines 2931-2933. Reduces the claim to one perfect object.
- The comparison map is an isomorphism for perfect objects. Same chapter, Lemma labelled lemma-compare-with-pullback-perfect, cited at proof lines 2933-2935. Settles the reduced case.

Source reference. Original equation 0A9S: sources/stacks-duality/pinned-original-source.tex, source lines 2079–2079. Same-chapter pointer into the author environment equation at source line 2079; invoked at proof line 2903 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0E4I: sources/stacks-duality/pinned-original-source.tex, source lines 2831–2831. Same-chapter pointer into the author environment lemma at source line 2831; invoked at proof line 2901 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original theorem 09M5: sources/stacks-chow/tags.txt, source lines 12168–12168. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2908 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original theorem 09IS: sources/stacks-chow/tags.txt, source lines 12050–12050. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2910 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 09PB: sources/stacks-chow/tags.txt, source lines 12244–12244. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2912 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9T: sources/stacks-duality/pinned-original-source.tex, source lines 2085–2085. Same-chapter pointer into the author environment lemma at source line 2085; invoked at proof line 2916 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let Y be a quasi-compact and quasi-separated scheme. Let $f : X \to Y$ be a morphism of schemes which is proper, flat, and of finite presentation. The map (0A9S) is an isomorphism for every object K of $D_{QCoh}(\mathcal{O}_Y)$.*

Proof. By Lemma 0E4I we know that a commutes with direct sums. Hence the collection of objects of $D_{QCoh}(\mathcal{O}_Y)$ for which (0A9S) is an isomorphism is a strictly full, saturated, triangulated subcategory of $D_{QCoh}(\mathcal{O}_Y)$ which is moreover preserved under taking direct sums. Since $D_{QCoh}(\mathcal{O}_Y)$ is a module category (Derived Categories of Schemes, Theorem 09M5) generated by a single perfect object (Derived Categories of Schemes, Theorem 09IS) we can argue as in More on Algebra, Remark 09PB to see that it suffices to prove (0A9S) is an isomorphism for a single perfect object. However, the result holds for perfect objects, see Lemma 0A9T. □